LLM Evaluation Case Study

Prompt Engineering & RLHF Portfolio

Case ID: [001] | Date: October 9, 2026

1. The Prompt

User Input:
Guess the next number: 7, 10, 16, 22, ?

2. Model Responses

Model A (claude-opus-5-max)

Those are the primes! So each term is $3p + 1$:

- $3(2) + 1 = 7$
- $3(3) + 1 = 10$
- $3(5) + 1 = 16$
- $3(7) + 1 = 22$
- $3(11) + 1 = \mathbf{34}$

Equivalently, the gaps $(3, 6, 6, 12)$ are just $3\times$ the prime gaps $(1, 2, 2, 4)$.

(Caveat: short sequences are never unique – if you intended the gaps to settle into a steady $+6$, the answer would be 28. But the prime-based rule explains all four given terms exactly, so 34 is my pick.)

Model B (gpt-5.6-terra-xhigh)

There's no unique answer from four terms alone. A simple continuation is **28**, using the repeating increments $+3, +6, +6$:

$7 \to 10\ (+3), \quad 10 \to 16\ (+6), \quad 16 \to 22\ (+6), \quad 22 \to 28\ (+6)$

But other rules could give different answers.

3. Evaluation & Reasoning

Decision: Model A
Category/Domain: Logical Reasoning & Pattern Recognition

Detailed Justification

- **Analytical Depth & Insight:** Model A identifies a highly robust, non-trivial mathematical relationship ($3p + 1$, mapping to the sequence of prime numbers). Model B, conversely, relies on a weak, almost arbitrary assumption that the increment simply starts repeating ($+3, +6, +6...$) after the third term.
- **Nuance & Anticipation:** Both models correctly note that short sequences are not unique. However, Model A exhibits exceptional reasoning by explicitly predicting Model B's output (28) as a potential secondary answer. By doing so, it preempts the simpler arithmetic progression while expertly justifying why its own prime-based rule is superior ("explains all four given terms exactly").
- **Tone & Formatting:** Model A uses a more engaging, insightful tone and presents the mathematical proof in a clean, readable list format.
- **Conclusion:** Model A is the definitive winner due to its significantly higher degree of logical deduction. It not only solves the problem with a deeper mathematical insight but also evaluates alternative simpler patterns within its own response.